



Laboratory Report #3

Laboratory Exercise 3: Simple Linear Regression Using JASP

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A. Instructions:

Scenario: As an engineering analyst, you're studying how engine load affects fuel consumption in a generator system. Use simple linear regression in JASP to model and interpret the relationship.

1. Analyze whether the 2 variables have a relationship.

r	Level of Correlation
0.00 - 0.25	Very Low
0.26 - 0.49	Low
0.50 – 0.69	Moderate
0.70 – 0.89	High
0.90 – 1.00	Very High

2. Open **JASP** and load the file engine_linear_regression.csv.
3. Go to **Regression** → **Linear Regression**.
4. Assign:
- **Dependent Variable:** Fuel_Consumption
 - **Covariate:** Engine_Load
5. Under **Statistics**, check:
- Model fit (R^2 , Adjusted R^2)
 - Confidence intervals
 - ANOVA table
6. Under **Plots**, enable:
- Residual plots
 - Predicted vs. Observed plot
7. Review the model summary:
- Regression equation: Intercept and slope
 - Significance (p-value)
 - R^2 value
8. Export your APA-style report and graphs as PDF.



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Tables

Linear Regression

Model Summary - Fuel_Consumption

Model	R	R ²	Adjusted R ²	RMSE
M ₀	0.000	0.000	0.000	1.216
M ₁	0.975	0.950	0.949	0.274

Note. M₁ includes Engine_Load

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M ₁	Regression	138.926	1	138.926	1850.555	< .001
	Residual	7.357	98	0.075		
	Total	146.283	99			

Note. M₁ includes Engine_Load

Note. The intercept model is omitted, as no meaningful information can be shown.

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p	95% CI	
							Lower	Upper
M ₀	(Intercept)	2.969	0.122		24.427	< .001	2.728	3.210
M ₁	(Intercept)	-0.063	0.076		-0.827	0.410	-0.213	0.088
	Engine_Load	0.051	0.001	0.975	43.018	< .001	0.048	0.053

Analysis:

1. What is the regression equation? Interpret the slope.

From the result of the Coefficients Table, which is the Model M₁, the regression equation for the Fuel-Consumption is $-0.063 + 0.051 (\text{Engine_Load})$. For every 1 unit increase in the engine load, the fuel consumption also increases by 0.051 units.

2. Is the model statistically significant?

Yes, the model is statistically significant because it satisfies the condition of p-value is less than 0.001, which can be shown from both the ANOVA and Coefficients table. This just means that the relationship between the engine load and fuel consumption is highly significant.

3. What does the R² value tell you about the strength of the prediction?

The R² value is 0.950 or 95%, which means that there is 95% variation in the fuel consumption that can be explained by the engine load.

4. Do the residual plots indicate a good model fit?

The residual plots drop from 1.216 (M₀) to 0.274 (M₁), which means that there is less likely an error to occur. The small residual errors also indicate or suggest that it is a good model fit

Video Link: [JARDELEZA_DSL_Laboratory3.mp4](#)